UP Board Class 12 Biology - 348 (CG) 2023 Question Paper with Solutions

Time Allowed :3 Hours	Maximum Marks :100	Total Questions :9

General Instructions

Read the following instructions very carefully and strictly follow them:

1. There are in all nine questions in this question paper.
2. All questions are compulsory.
3. In the beginning of each question, the number of parts to be attempted are clearly mentioned.
4. Marks allotted to the questions are indicated against them.
5. Start solving from the first question and proceed to solve till the last one. Do not waste your time over a question you cannot solve.

1.(a) In Vallisneria, the pollination occurs by means of

(A) Water
(B) Air
(C) Insects
(D) Human beings

Correct Answer: (A) Water

Solution:

Step 1: Understanding Vallisneria.
Vallisneria is an aquatic plant that depends on water for pollination. Its male flowers are released into the water and carried to the submerged female flowers. This type of pollination is called hydrophily.

Step 2: Analysis of options.
- (A) Water: Correct, as Vallisneria uses water for pollination.
- (B) Air: Incorrect, air pollination occurs in terrestrial plants, not aquatic ones like Vallisneria.
- (C) Insects: Incorrect, Vallisneria does not rely on insects for pollination.
- (D) Human beings: Incorrect, humans do not play a role in Vallisneria's pollination.

Step 3: Conclusion.
Thus, the correct answer is (A) Water, since Vallisneria is pollinated through water.

Quick Tip

Hydrophily is the term used for water pollination, which is common in aquatic plants like Vallisneria.

(b) Menstrual cycle in women typically occurs in

(A) 10-11 days
(B) 28-29 days
(C) 25-26 days
(D) 17 days

Correct Answer: (B) 28-29 days

Solution:

Step 1: Understanding the Menstrual Cycle.
The menstrual cycle in women is a regular natural process that prepares the body for pregnancy. It generally lasts around 28 days, though it can vary between 21 and 35 days. The cycle consists of several phases: menstrual, follicular, ovulation, and luteal.

Step 2: Analysis of options.
- (A) 10-11 days: This duration is too short to be a typical menstrual cycle.
- (B) 28-29 days: This represents the average and most common length of the menstrual cycle.
- (C) 25-26 days: Slightly shorter than average but still within a normal range.
- (D) 17 days: This is shorter than the usual duration for a menstrual cycle.

Step 3: Conclusion.
Therefore, the typical menstrual cycle length is 28-29 days, making option (B) the correct choice.

Quick Tip

The average menstrual cycle is typically 28 days, though it can vary between 21 and 35 days in some women.

(c) The process of protein synthesis with the help of m-RNA is called

(A) Translation
(B) Transcription
(C) Reverse transcription
(D) Replication

Correct Answer: (A) Translation

Solution:

Step 1: Understanding the Process of Protein Synthesis.
Protein synthesis occurs in two main stages: transcription and translation. During transcription, mRNA is produced from the DNA template, while translation uses this mRNA to assemble proteins.

Step 2: Analysis of options.
- (A) Translation: This is the stage where proteins are synthesized using mRNA, so it is the correct answer.
- (B) Transcription: This stage involves the formation of mRNA from DNA, not the synthesis of proteins.
- (C) Reverse transcription: This process converts RNA back into DNA, occurring mainly in certain viruses, and is unrelated to normal protein synthesis.
- (D) Replication: This is the copying of DNA molecules, not related to protein synthesis.

Step 3: Conclusion.
Thus, the stage of protein synthesis involving mRNA is called Translation, making option (A) correct.

Quick Tip

Translation is the process where mRNA is decoded to synthesize proteins in the ribosome.

(d) Widal test is used for the diagnosis of which disease?

(A) Pneumonia
(B) Typhoid
(C) Malaria
(D) Cholera

Correct Answer: (B) Typhoid

Solution:

Step 1: Understanding the Widal Test.
The Widal test is a serological test used to detect antibodies against *Salmonella typhi*, the bacterium responsible for typhoid fever. It is widely used in clinical settings for the diagnosis of typhoid.

Step 2: Analysis of options.
- (A) Pneumonia: Caused by various pathogens such as bacteria, viruses, and fungi, but not *Salmonella typhi*. The Widal test is not applicable here.
- (B) Typhoid: Caused by *Salmonella typhi*, and the Widal test specifically detects this infection. Hence, this is the correct option.
- (C) Malaria: Caused by *Plasmodium* species and diagnosed by blood smear or rapid tests, not by the Widal test.
- (D) Cholera: Caused by *Vibrio cholerae*, unrelated to *Salmonella typhi*, so the Widal test is not used.

Step 3: Conclusion.
Therefore, the Widal test is used to diagnose typhoid fever, making option (B) the correct answer.

Quick Tip

The Widal test is primarily used for diagnosing typhoid fever caused by *Salmonella typhi*.

2.(a) Define the biodiversity hotspot.

Solution:

A **biodiversity hotspot** is a biogeographic region that serves as a significant reservoir of biodiversity while simultaneously facing severe threats from human activities. These areas are characterized by a high concentration of endemic species—species found nowhere else on Earth—and are experiencing extensive habitat loss. The term was first coined by Norman Myers in 1988 and has become fundamental in conservation biology.
According to Conservation International, a region qualifies as a biodiversity hotspot if it meets the following criteria:

1. **Endemism Criterion:** The region must contain at least 1,500 species of vascular plants that are endemic.
2. **Threatened Habitat Criterion:** The region must have lost at least 70% of its original natural vegetation.

These hotspots are vital for biodiversity conservation as they cover only about 2.3% of the Earth's land surface but harbor more than half of the world's endemic plant species and a large proportion of animal species.

Examples of Biodiversity Hotspots:

- **The Amazon Rainforest:** Located in South America, it is one of the richest ecosystems in the world, home to countless species, many still undiscovered. Threats include deforestation and climate change.

- **Sundaland:** Encompassing regions like Borneo and Sumatra in Southeast Asia, this hotspot is notable for species such as the orangutan and Sumatran tiger. Major threats include deforestation and illegal wildlife trade.

Quick Tip

Biodiversity hotspots are often small in size but have a large proportion of the world's species. Protecting them is essential for preserving global biodiversity. These areas are often at risk due to human activities like deforestation and urbanization.

(b) Which vitamin is abundantly found in Golden rice?

Solution:
Golden rice is a genetically modified (GM) variety of rice that has been engineered to produce high levels of **Vitamin A**, specifically in the form of **beta-carotene**. The primary goal of developing Golden rice was to address the widespread issue of Vitamin A deficiency (VAD), particularly in developing countries where rice is a staple food.
What is Vitamin A and Why is it Important?
Vitamin A is a fat-soluble vitamin that plays an essential role in several physiological functions:

1. **Vision:** Vitamin A is a key component of rhodopsin, a protein in the eyes that helps with vision in low-light conditions.
2. **Immune System:** It supports the immune system by maintaining the integrity of the skin and mucous membranes, which act as barriers to pathogens.
3. **Cell Growth and Reproduction:** Vitamin A is vital for normal growth and development, particularly in the fetus during pregnancy.
4. **Skin Health:** It helps maintain healthy skin and epithelial tissues, and is used in the treatment of various skin conditions.

Beta-Carotene in Golden Rice:
Beta-carotene is a precursor to Vitamin A. This means that the body can convert beta-carotene into Vitamin A as needed. Golden rice contains high amounts of beta-carotene due to the insertion of specific genes responsible for the biosynthesis of carotenoids. These genes were sourced from the daffodil flower and a bacterium (Erwinia uredovora) to enable the rice plant to produce carotenoids in its grains, which would otherwise be absent.
Global Health Impact of Golden Rice:
Vitamin A deficiency is a leading cause of preventable blindness in children and contributes to increased mortality due to compromised immune systems. It is especially prevalent in developing nations where access to a diverse diet is limited. Golden rice was developed to provide an affordable and sustainable source of Vitamin A to populations at risk of VAD, particularly in areas of Asia where rice is a staple food.
The World Health Organization (WHO) has endorsed Golden rice as a potential intervention to combat VAD in these regions. However, it is still undergoing regulatory approvals in several

countries, and its widespread adoption has been slowed by debates surrounding genetically modified organisms (GMOs).

Quick Tip

Golden rice provides a source of Vitamin A in regions where the diet is deficient in this nutrient. Beta-carotene in Golden rice helps prevent Vitamin A deficiency, which can lead to blindness and increased mortality in children.

(c) What is the function of Sertoli cells in human testes?

Solution:
Sertoli cells, also known as "nurse cells," are specialized cells found in the seminiferous tubules of the testes. These cells are essential for the process of spermatogenesis, which is the production of sperm. Their functions include the following:

1. Support and nourishment of developing sperm cells:
Sertoli cells provide physical support and nourishment to the developing spermatozoa throughout spermatogenesis. They provide a conducive environment for the growth and maturation of sperm cells from spermatogonia (stem cells) to mature spermatozoa.

2. Formation of the Blood-Testis Barrier:
Sertoli cells create the blood-testis barrier, a specialized structure that prevents harmful substances and immune cells from accessing the developing sperm cells. This barrier is formed by tight junctions between adjacent Sertoli cells, ensuring that only necessary substances reach the sperm cells. It also protects sperm from immune system attack, as sperm cells are genetically different from the body and could be recognized as foreign invaders.

3. Secretion of Hormones:
Sertoli cells secrete various hormones and proteins essential for the regulation of spermatogenesis. One of the key hormones secreted is *Inhibin*, which regulates the production of Follicle Stimulating Hormone (FSH) by the pituitary gland, thereby controlling the rate of spermatogenesis. They also secrete *androgen-binding protein (ABP)*, which binds to testosterone, ensuring that the testosterone concentration remains high within the seminiferous tubules, further supporting sperm maturation.

4. Phagocytosis of Residual Cytoplasm:
During spermatogenesis, as the spermatid transforms into a mature spermatozoon, excess cytoplasm is eliminated. Sertoli cells play a crucial role in phagocytizing (engulfing and digesting) this excess cytoplasm, ensuring the formation of a compact and mature sperm cell.

5. Regulation of Spermatogenesis:
Sertoli cells also help regulate the process of spermatogenesis by responding to hormonal signals from the pituitary and hypothalamus. They communicate with germ cells through paracrine

signaling to support their development.

Quick Tip

Sertoli cells are often referred to as "nurse cells" because they support and regulate the process of sperm production and maturation in the testes.

(d) How many pairs of chromosomes are found in human beings?

Solution:
Human beings have a total of 46 chromosomes, which are divided into 23 pairs. These chromosomes contain the genetic material that determines the characteristics of an individual. The pairs are divided into two categories: autosomes and sex chromosomes.

1. Autosomes:
Out of the 23 pairs, 22 are autosomes. These are the non-sex chromosomes and are numbered from 1 to 22. They carry genes that are responsible for the majority of an individual's traits, such as eye color, height, and blood type. The autosomes are the same in both males and females.

2. Sex Chromosomes:
The remaining pair consists of sex chromosomes, which determine the biological sex of the individual. There are two types of sex chromosomes: X and Y.
- In females, the sex chromosomes are XX.
- In males, the sex chromosomes are XY.

The presence of two X chromosomes (XX) typically results in a female, while one X and one Y chromosome (XY) results in a male. The sex chromosomes not only determine sex but also carry genes that influence sexual development and fertility.

3. Chromosomal Disorders:
The number of chromosomes is usually fixed at 46 (23 pairs). However, in some cases, there can be chromosomal abnormalities, such as Down syndrome, where an individual has 47 chromosomes (an extra copy of chromosome 21). Other disorders like Turner syndrome (45 chromosomes, missing one X in females) and Klinefelter syndrome (47 chromosomes, with an extra X in males) can also occur due to chromosomal abnormalities.

Thus, the total number of pairs of chromosomes in humans is 23.

Quick Tip

Humans have 46 chromosomes arranged into 23 pairs: 22 pairs of autosomes and 1 pair of sex chromosomes (XX or XY).

(e) Who discovered Penicillin?

Solution:

Penicillin was discovered by **Alexander Fleming** in 1928. The story of its discovery is both accidental and revolutionary. Fleming, a Scottish bacteriologist, was working at St. Mary's Hospital in London when he made this groundbreaking observation. One day, while examining bacterial cultures, he noticed a mold growing on a Petri dish that had been accidentally left uncovered. Around the mold, there was a clear area where the bacteria were unable to grow. This led Fleming to conclude that the mold was producing a substance that inhibited the growth of bacteria.

The mold was later identified as *Penicillium notatum*, and the substance it produced was named **penicillin**. Fleming's initial experiments showed that penicillin was effective in killing a wide variety of bacteria, including those that caused infections like pneumonia, scarlet fever, and gonorrhea. However, it was not immediately developed for medical use.

Fleming's discovery was initially met with skepticism, but it was further developed in the 1940s by scientists like Howard Florey, Sir Norman Heatley, and Sir Alexander Fleming himself. Their collaborative work led to the mass production of penicillin during World War II, where it proved essential in treating bacterial infections among soldiers, ultimately saving millions of lives. It was the first widely available antibiotic, and it marked the beginning of the antibiotic era.

Impact of Penicillin:
Penicillin revolutionized the field of medicine. Before its discovery, many bacterial infections were often fatal. Diseases like pneumonia, tuberculosis, and septicemia were leading causes of death. The advent of penicillin, and later other antibiotics, transformed medicine, allowing doctors to treat and cure infections that were previously deadly. Penicillin's discovery also paved the way for the development of other antibiotics, which has helped in the fight against infectious diseases worldwide.

Fleming's work earned him the Nobel Prize in Physiology or Medicine in 1945, shared with Florey and Chain for their role in the development of penicillin. His discovery is often regarded as one of the most important medical breakthroughs of the 20th century.

Quick Tip

Fleming's discovery was accidental, but it led to a major shift in how bacterial infections are treated, saving millions of lives. The development of penicillin marked the beginning of the antibiotic era.

3.(a) What do you mean by medical termination of pregnancy (MTP)?

Solution:

Medical Termination of Pregnancy (MTP):
Medical Termination of Pregnancy (MTP) refers to the process of terminating a pregnancy by medical methods. It is typically performed within the first 12 weeks of pregnancy, although in certain cases, it may be allowed up to 24 weeks, depending on the medical guidelines and reasons. MTP can be done through medication or surgical procedures.

Methods of MTP:
1. **Medical methods:** These involve the use of medications such as Mifepristone and Misoprostol to induce abortion. Mifepristone blocks the hormone progesterone, which is necessary for pregnancy maintenance, and Misoprostol causes the uterus to contract and expel the pregnancy.

2. **Surgical methods:** If the medical method is not suitable or if the pregnancy is beyond 12 weeks, surgical procedures such as aspiration (suction) abortion or dilation and curettage (D&C) may be used.

Reasons for Medical Termination:
1. **Health risks to the mother:** If continuing the pregnancy poses a danger to the mother's physical or mental health.

2. **Fetal abnormalities:** If the fetus has congenital or genetic abnormalities that would lead to death after birth or result in a life of suffering.

3. **Unintended pregnancy:** Women may choose MTP if the pregnancy is unplanned and not desired due to personal or socioeconomic reasons.

Legal Aspects:
In many countries, medical termination of pregnancy is legal under specific conditions. The Medical Termination of Pregnancy Act, 1971 (in India) allows for the termination of pregnancy by a registered medical practitioner under certain conditions, such as the risk to the mother's life or physical health, or if the fetus has a serious abnormality.

(b) If the sequence of Nitrogenous bases in one strand of DNA-helix is 5'-ATGCATGCATGCA 3' then what will be the sequence of nitrogenous bases in the complementary strand of that DNA helix in 3' → 5' direction?

Solution:

Base Pairing in DNA:
In DNA, the two strands are complementary to each other, meaning that the nitrogenous bases

on one strand pair with complementary bases on the opposite strand. The base pairing rules are as follows: - **Adenine (A) pairs with Thymine (T)**
- **Thymine (T) pairs with Adenine (A)**
- **Cytosine (C) pairs with Guanine (G)**
- **Guanine (G) pairs with Cytosine (C)**

Step 1: Understanding the given DNA sequence The given sequence of one strand of DNA is:

$$5' - \text{ATGCATGCATGCATGC} - 3'$$

Step 2: Writing the complementary bases Now, we need to find the complementary bases for each nucleotide in the sequence. Applying the base pairing rules, we get:

$$5' - A \,\text{pairs with T}, \quad T \,\text{pairs with A}, \quad G \,\text{pairs with C}, \quad C \,\text{pairs with G}$$

Step 3: Complementary sequence in the 3' to 5' direction Starting from the left end (5') of the given strand, the complementary sequence will be formed as follows:
- 5'-A T-3'
- 5'-T A-3'
- 5'-G C-3'
- 5'-C G-3'
- 5'-A T-3'
- 5'-T A-3'
- 5'-G C-3'
- 5'-C G-3'
- 5'-A T-3'
- 5'-T A-3'
- 5'-G C-3'
- 5'-C G-3'

Thus, the complementary sequence in the 3' $\rightarrow$ 5' direction will be:

$$3' - \text{TACGTACGTACGTACG} - 5'$$

Step 4: Final answer The sequence of nitrogenous bases in the complementary strand in the 3' $\rightarrow$ 5' direction is:

$$\boxed{3' - TACGTACGTACGTACG - 5'}$$

Quick Tip

When working with DNA sequences, remember that **adenine (A) pairs with thymine (T)**, and **cytosine (C) pairs with guanine (G)**. The complementary strand is always read in the 3' to 5' direction.

(c) Draw only labelled diagram of a single antibody molecule.

Solution:
An antibody molecule, also known as an immunoglobulin (Ig), consists of four polypeptide chains: two heavy chains (H) and two light chains (L). The structure of an antibody is typically Y-shaped, with the arms of the Y representing the antigen-binding sites.
The main parts of an antibody are:

1. **Antigen-binding site:** The part of the antibody that binds to a specific antigen. This is located at the tips of the Y-shaped structure.
2. **Variable region:** The portion of the antibody that varies to recognize different antigens.
3. **Constant region:** The region that is identical for all antibodies of the same class. It determines the mechanism used to destroy the antigen.
4. **Heavy chains (H):** The larger polypeptide chains that form the base of the Y and contribute to the constant region.
5. **Light chains (L):** The smaller polypeptide chains attached to the heavy chains that form the arms of the Y.
6. **Fc region:** The tail of the Y-shaped antibody that binds to receptors on immune cells.

Quick Tip

Antibodies are highly specific proteins that bind to antigens. The diversity in antibodies is created by the variability in the antigen-binding sites.

(d) What is Heroin (Morphine)? This is obtained from which plant?

Solution:
Heroin and morphine are both powerful opioids that are derived from the opium poppy plant, *Papaver somniferum*. The opium poppy has been cultivated for thousands of years for its medicinal and recreational uses.
Morphine:
Morphine is a naturally occurring alkaloid extracted from the opium poppy. It is used as a pain reliever, especially in severe cases, as it has potent analgesic (pain-relieving) properties. Morphine works by binding to opioid receptors in the brain and spinal cord, which blocks pain signals and induces feelings of euphoria.
Heroin:
Heroin, also known as diacetylmorphine, is a semi-synthetic opioid derived from morphine. Heroin is synthesized by adding two acetyl groups to the morphine molecule, which makes it more lipophilic and allows it to cross the blood-brain barrier more quickly. Heroin is highly addictive and illegal in many parts of the world due to its dangerous potential for abuse and overdose.
How Heroin is Obtained:
The process begins with the cultivation of the opium poppy, which produces a milky latex substance containing alkaloids, including morphine. The latex is collected and refined to obtain

pure morphine. From morphine, heroin is synthesized by chemical processes. This transformation significantly enhances its potency, making heroin much more addictive and dangerous than morphine.

Quick Tip

Heroin is a derivative of morphine, both of which are extracted from the opium poppy. Heroin is much more potent and addictive than morphine.

(e) What do you mean by central dogma of Molecular Biology?

Solution:
The central dogma of molecular biology is a fundamental concept that describes the flow of genetic information within a biological system. It was first proposed by Francis Crick in 1958 and outlines the process by which genetic information is transferred from DNA to RNA and finally to proteins. This process is crucial for cellular function and organism development. The central dogma can be summarized as follows:

$$\text{DNA} \xrightarrow{\text{Transcription}} \text{mRNA} \xrightarrow{\text{Translation}} \text{Protein}$$

Explanation of the Process:
1. **Transcription:** The process begins with the transcription of DNA into messenger RNA (mRNA). In this step, a segment of DNA is copied into mRNA by the enzyme RNA polymerase. The mRNA serves as a blueprint that carries genetic information from the DNA in the nucleus to the cytoplasm, where proteins are synthesized.

2. **Translation:** In the cytoplasm, the mRNA is translated into a protein. Ribosomes read the sequence of the mRNA in sets of three nucleotides, called codons, which specify a particular amino acid. Transfer RNA (tRNA) molecules bring the appropriate amino acids, which are linked together to form a protein. The sequence of amino acids determines the structure and function of the protein.

3. **Role of Proteins:** Proteins are essential for carrying out various cellular functions, including catalyzing metabolic reactions (enzymes), providing structural support (cytoskeleton), and regulating cellular activities (hormones, antibodies). Proteins are the workhorses of the cell, executing the instructions encoded in the DNA.

Exceptions and Extensions:
While the central dogma primarily describes the flow of information from DNA to RNA to protein, there are some exceptions and extensions. For example, some viruses (like retroviruses) can reverse this flow, converting RNA into DNA via an enzyme called reverse transcriptase. This phenomenon is known as reverse transcription.

Quick Tip

The central dogma explains the essential process of how genetic information is converted into functional proteins. Remember, it's the flow from DNA to RNA to protein for most biological systems.

4.(a) Write short notes on Lichen and Mycorrhiza.

Solution:
Lichen:
Lichens are composite organisms that arise from the symbiotic association between a fungus and a photosynthetic partner, which is typically an alga or a cyanobacterium. This mutualistic relationship benefits both partners: the fungus provides protection and moisture to the photosynthetic partner, while the photosynthetic partner provides food (carbohydrates) to the fungus. Lichens are important indicators of environmental health, especially air quality, because they are sensitive to pollution, particularly sulfur dioxide. They can grow on a variety of surfaces, including rocks, tree bark, and soil.
Types of Lichens:
1. **Crustose:** Lichens that form a crust-like appearance on the surface of the substrate.
2. **Foliose:** Leafy lichens that are loosely attached to the substrate.
3. **Fruticose:** Bushy lichens that have a more three-dimensional structure.

Mycorrhiza:
Mycorrhiza refers to a symbiotic association between fungi and the roots of plants. This relationship is beneficial for both organisms: the fungus assists the plant in nutrient uptake, especially phosphorus, while the plant provides the fungus with carbohydrates produced through photosynthesis. Mycorrhiza is crucial for the health and growth of many plants, as it helps in nutrient absorption and increases the plant's resistance to pathogens and environmental stress.
Types of Mycorrhiza:
1. **Ectomycorrhiza:** The fungal hyphae form a dense layer around the root surface but do not penetrate the cells. Common in woody plants like pines and oaks. 2. **Endomycorrhiza:** The fungal hyphae penetrate the plant's root cells, forming structures such as arbuscules. Common in herbaceous plants and many crop species.

Quick Tip

Lichens are an important ecological indicator, while mycorrhizal fungi enhance plant growth by aiding nutrient uptake, especially phosphorus.

(b) What is codominance? Explain with an example.

Solution:

Codominance:
Codominance is a genetic phenomenon in which two different alleles of a gene are both expressed in the phenotype of an organism. Unlike incomplete dominance, where one allele partially masks the expression of the other, in codominance, both alleles contribute equally to the organism's traits. This results in a heterozygous individual displaying both traits simultaneously.
Example of Codominance:
A well-known example of codominance is the inheritance of the **AB blood group** in humans. The A and B alleles are codominant, meaning that if an individual inherits both the A allele from one parent and the B allele from the other parent, the resulting phenotype will express both A and B antigens on the surface of the red blood cells. This results in the AB blood type.
Genotype and Phenotype:
- **Genotype:** IAIB (heterozygous)
- **Phenotype:** AB blood type

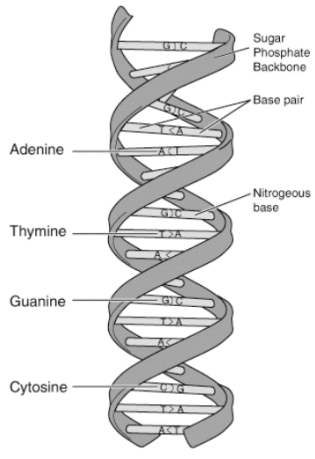
In this case, both the A and B alleles are fully expressed, making it a clear example of codominance.

Quick Tip

Codominance occurs when both alleles contribute equally to the organism's phenotype, as seen in the AB blood group.

(c) Draw only labelled diagram of a DNA double helix structure.

Solution:
A labelled diagram of the DNA double helix structure is as follows:



Key Features of the DNA Double Helix Structure:
1. **Double Strands**: DNA consists of two long chains of nucleotides coiled around each other, forming a double helix structure.
2. **Sugar-Phosphate Backbone**: The backbone of each strand is made up of alternating sugar (deoxyribose) and phosphate groups.

3. **Nitrogenous Bases**: Each strand contains nitrogenous bases (Adenine, Thymine, Cytosine, Guanine) that pair specifically:
- Adenine (A) pairs with Thymine (T) via two hydrogen bonds.
- Cytosine (C) pairs with Guanine (G) via three hydrogen bonds.
4. **Complementary Base Pairing**: The two strands of DNA are complementary, meaning the sequence of bases on one strand dictates the sequence on the other strand.

> Quick Tip
>
> In the DNA double helix, base pairing ensures that the genetic information is stored accurately. A pairs with T, and C pairs with G.

(d) What do you understand by transcription? Describe it.

Solution:
Transcription is the process by which an RNA molecule is synthesized from a DNA template. It is the first step in the central dogma of molecular biology, where genetic information in DNA is used to produce a complementary RNA strand. The process of transcription occurs in three main stages:

1. Initiation:
The process begins when the enzyme RNA polymerase binds to the promoter region of the DNA. The promoter is a specific sequence of nucleotides that signals the start of a gene. RNA polymerase unwinds the DNA strands, exposing the template strand.

2. Elongation:
RNA polymerase moves along the DNA template strand, reading the sequence of nucleotides and synthesizing a complementary RNA strand. In this stage, RNA is built in the 5' to 3' direction, with RNA nucleotides (A, U, C, G) pairing with the complementary bases on the DNA strand (A pairs with U, and C pairs with G).

3. Termination:
Once RNA polymerase reaches a specific sequence called the terminator, the RNA transcript is released. The RNA molecule then detaches from the DNA, and the DNA strands re-anneal. The newly formed RNA molecule is called messenger RNA (mRNA) in eukaryotes and can undergo further processing before translation.

Key Points:
- Transcription is essential for protein synthesis, as it produces the mRNA, which carries the genetic code from DNA to ribosomes for translation.
- In eukaryotic cells, transcription occurs in the nucleus, while in prokaryotes, it occurs in the cytoplasm.

Quick Tip

Transcription is the first step in gene expression, where DNA is transcribed into mRNA. Remember, in RNA, uracil (U) replaces thymine (T).

5.(a) What is endosperm? Describe the process of its development and functions.

Solution:

The **endosperm** is a tissue found in the seeds of angiosperms (flowering plants). It is formed during fertilization and serves as a food reserve for the developing embryo. The endosperm is triploid (having three sets of chromosomes) in most species, resulting from the fusion of two polar nuclei with a sperm cell during the process of double fertilization.

Process of Development:
Endosperm development occurs following fertilization. After the sperm cell fuses with the egg cell to form the zygote, another sperm cell fuses with the two polar nuclei to form the triploid endosperm. This endosperm then starts dividing and growing to form a mass of tissue, which provides nourishment to the developing embryo within the seed. Depending on the species, the endosperm may develop in different ways:
- In **monocots**, the endosperm persists in the mature seed.
- In **dicots**, the endosperm is often absorbed by the embryo during seed development.

Functions:
1. The endosperm provides essential nutrients (such as starches, proteins, and fats) to the developing embryo, supporting early growth after seed germination.
2. It acts as a storage organ in many seeds, ensuring that the seedling has enough energy for initial growth before it can photosynthesize.
3. In some plants, the endosperm plays a role in protecting the seed from environmental stresses.

Final Answer: Endosperm is a crucial tissue in seed development, providing nourishment for the embryo and helping in early seedling growth.

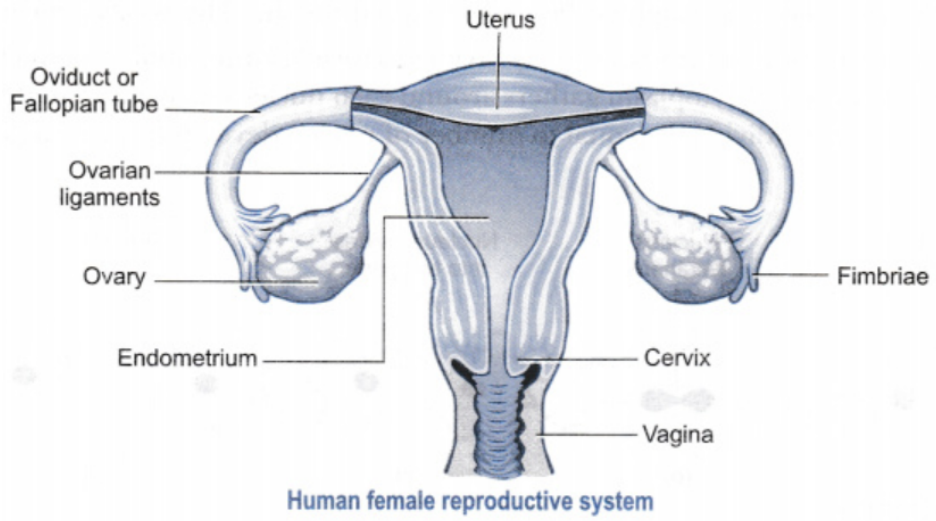
Quick Tip

Endosperm development occurs through double fertilization, and its primary function is to nourish the developing embryo.

(b) Draw a labelled line diagram of women's reproductive system.

Solution:

Below is a labelled line diagram of the female reproductive system:



Human female reproductive system

The diagram should include the following parts:
- **Ovary**: The female gonads that produce eggs (ova) and secrete hormones (estrogen and progesterone).
- **Fallopian Tube (Oviduct)**: The tube through which eggs travel from the ovaries to the uterus. Fertilization typically occurs here.
- **Uterus**: The organ where the fertilized egg implants and develops into a fetus.
- **Cervix**: The lower part of the uterus that connects to the vagina.
- **Vagina**: The muscular canal that leads from the cervix to the outside of the body, serving as the birth canal during childbirth.

Quick Tip

The female reproductive system is designed to produce eggs, facilitate fertilization, and support fetal development.

(c) Define sex linked inheritance. Explain it with haemophilia and colourblindness.

Solution:

Sex-linked inheritance:
Sex-linked inheritance refers to the inheritance pattern of genes located on the sex chromosomes, typically the X chromosome. In humans, males have one X and one Y chromosome

(XY), whereas females have two X chromosomes (XX). This difference results in different inheritance patterns for males and females. Males are more often affected by sex-linked disorders because they have only one X chromosome, while females have a second X chromosome to potentially compensate for the defective gene.

Haemophilia:
Haemophilia is a sex-linked recessive disorder. The gene responsible for haemophilia is located on the X chromosome. Males with the defective gene on their single X chromosome will express the disease because they have no second X chromosome to mask the defective gene. Females, on the other hand, would need to inherit the defective gene on both X chromosomes to exhibit the disease. If a female inherits the defective gene on only one X chromosome, she becomes a carrier and does not show symptoms of the disease.

Colourblindness:
Colourblindness is another example of a sex-linked recessive disorder. The gene for colour vision is located on the X chromosome. Like haemophilia, males with the defective gene on their single X chromosome will be colourblind. Females need to inherit two copies of the defective gene (one from each parent) to be colourblind. A female with only one defective copy will be a carrier but not affected by the condition.

Quick Tip

In sex-linked inheritance, traits are passed through the X chromosome. Males are more often affected because they have only one X chromosome, while females have two, which can compensate for one defective X.

(d) Comment upon PCR.

Solution:

Polymerase Chain Reaction (PCR):
Polymerase Chain Reaction (PCR) is a revolutionary laboratory technique used to amplify a specific segment of DNA. Developed by Kary Mullis in 1983, PCR allows scientists to produce millions of copies of a particular DNA segment in a short period. The process involves the following steps:

1. **Denaturation:** The DNA sample is heated to around 94°C to separate the double-stranded DNA into two single strands.
2. **Annealing:** The reaction temperature is lowered to around 50–60°C, allowing short DNA primers to bind to the complementary sequences on the single-stranded DNA.
3. **Extension:** The temperature is increased to around 72°C, and a heat-stable DNA polymerase enzyme (such as Taq polymerase) is used to synthesize a new DNA strand complementary to the template strand.

PCR is widely used in various applications such as DNA cloning, gene analysis, forensics, and medical diagnostics. It has become an essential tool in modern molecular biology.

> Quick Tip
>
> Remember that PCR amplifies a specific segment of DNA, and it involves repeated cycles of denaturation, annealing, and extension. It's widely used in forensics, cloning, and diagnostics.

6.(a) Describe the Mendel's law of dominance.

Solution:
Mendel's Law of Dominance is one of the three fundamental principles of inheritance that were established by Gregor Mendel in the 19th century. According to Mendel's law of dominance, when an organism inherits two alleles for a particular trait, one allele may be dominant and the other recessive. The dominant allele will mask the expression of the recessive allele in the organism's phenotype, meaning that the trait controlled by the dominant allele will be expressed, while the trait controlled by the recessive allele will not.
Key Terms:

1. **Dominant Allele:** The allele that expresses its trait even in the presence of a different allele. It is usually represented by a capital letter (e.g., T for tall in pea plants).
2. **Recessive Allele:** The allele that is masked in the presence of a dominant allele. It only expresses its trait when an organism has two copies of the recessive allele. It is represented by a lowercase letter (e.g., t for short in pea plants).
3. **Homozygous:** When an organism inherits two identical alleles for a trait (e.g., TT or tt).
4. **Heterozygous:** When an organism inherits two different alleles for a trait (e.g., Tt).

Explanation with Example: Consider the example of pea plants, which were used by Mendel in his experiments. In this case, the height of the plant is governed by a single gene with two alleles: - The dominant allele (T) causes tall plants. - The recessive allele (t) causes short plants.
Mendel crossbred plants with different genotypes and observed the following:
- **Homozygous Dominant (TT):** These plants were tall because the dominant allele T was expressed. - **Homozygous Recessive (tt):** These plants were short because both alleles were recessive, and no dominant T allele was present to mask the effect. - **Heterozygous (Tt):** These plants were also tall because the dominant allele T masked the effect of the recessive allele t.
Mendel's First Experiment (Monohybrid Cross): Mendel crossed a homozygous tall plant (TT) with a homozygous short plant (tt). All the F1 offspring in this cross had the genotype Tt and

exhibited the dominant tall phenotype. This demonstrated that the dominant allele (T) masks the recessive allele (t) in the heterozygous state.
When Mendel crossed two F1 plants (Tt), the resulting F2 generation had a 3:1 phenotypic ratio, with 75% of the plants being tall (dominant phenotype) and 25% being short (recessive phenotype). This result confirmed the principle of dominance and suggested that the alleles segregate independently during gamete formation.

Quick Tip

The dominant allele expresses its trait in both homozygous (TT) and heterozygous (Tt) conditions, while the recessive allele only expresses its trait in the homozygous recessive condition (tt).

(b) What do you mean by polygenic inheritance? Describe with suitable example.

Solution:
Polygenic Inheritance:
Polygenic inheritance refers to the type of inheritance in which a characteristic is determined by the combined effects of two or more genes, and these genes are often located at different loci on different chromosomes. Unlike Mendelian inheritance, where a single gene controls a trait with distinct dominant and recessive alleles, polygenic traits exhibit a continuous range of phenotypes. These traits are often quantitative and show a gradual change, rather than the discrete classes seen in Mendel's experiments.
In polygenic inheritance, each gene contributes a small, additive effect to the overall phenotype. As a result, the phenotype does not follow the simple dominant/recessive pattern, and the expression of the trait can vary along a spectrum. This continuous variation leads to a wide range of phenotypic expressions, such as height, skin color, and weight.
Characteristics of Polygenic Traits: 1. **Multiple Genes Involved:** More than one gene controls the trait. 2. **Continuous Variation:** The trait shows a range of phenotypes, rather than discrete categories. 3. **Additive Effect:** Each gene contributes additively to the overall phenotype, so the more dominant alleles present, the more pronounced the trait.
Example of Polygenic Inheritance: **Human Skin Color:** Human skin color is a classic example of polygenic inheritance. It is controlled by the interaction of several genes, each contributing to the overall pigmentation of the skin. The primary genes involved are located on different chromosomes, and each gene has multiple alleles that contribute to skin pigmentation.
- Individuals with darker skin have more of the dominant alleles for pigmentation, leading to higher amounts of melanin. - Individuals with lighter skin have a greater number of recessive alleles for pigmentation, resulting in less melanin production.
The combination of alleles from each gene creates a continuous spectrum of skin tones, from very light to very dark.
Other Examples of Polygenic Inheritance: - **Human Height:** Height is determined by the interaction of multiple genes. While each gene has a small effect on an individual's height, the overall combination of alleles determines whether a person is short, average, or tall. - **Intelligence:** Like height and skin color, intelligence is influenced by multiple genes, although

environmental factors also play a significant role.
Mendel's Model vs. Polygenic Inheritance: Unlike Mendel's experiments, where traits were controlled by single genes with clear dominant and recessive alleles, polygenic inheritance involves multiple genes that interact to produce a range of phenotypes. Therefore, polygenic inheritance patterns do not follow the simple Mendelian ratios of 3:1 or 1:2:1 but rather produce bell-shaped curves when the phenotypic distribution is plotted.

Quick Tip

Polygenic traits are influenced by multiple genes, each contributing a small amount to the overall phenotype, resulting in continuous variation.

(c) Describe the main characteristic features of Genetic Code.

Solution:
The genetic code refers to the set of rules by which information encoded in DNA (or RNA) is translated into proteins. The main characteristic features of the genetic code are as follows:

1. Triplet Code:
Each amino acid in a protein is specified by a sequence of three nucleotide bases, known as a codon. The codons are read in sets of three from the mRNA during the process of translation.

2. Universality:
The genetic code is nearly universal. It is the same in almost all organisms, from bacteria to humans. This universality reflects the evolutionary common ancestry of all life forms.

3. Degeneracy (Redundancy):
The genetic code is degenerate, meaning that more than one codon can code for the same amino acid. For example, both UUU and UUC codons code for the amino acid phenylalanine.

4. Unambiguous:
Each codon specifies only one amino acid. There is no ambiguity in the interpretation of codons in the genetic code.

5. Start and Stop Codons:
The genetic code includes special codons known as the start codon (AUG), which signals the beginning of translation, and stop codons (UAA, UAG, UGA), which signal the end of translation.

6. Non-overlapping:
The genetic code is read in a non-overlapping manner, meaning that each nucleotide is part of only one codon, and codons do not overlap with each other.

7. Directionality:
The genetic code is read in the 5' to 3' direction during transcription and translation. This

directionality is essential for the proper synthesis of proteins.

> Quick Tip
>
> Remember, the genetic code is a universal, degenerate triplet code that is read without ambiguity and is essential for the synthesis of proteins.

(d) What are the causal organisms of typhoid and pneumonia? Give their symptoms.

Solution:

1. Typhoid:
The causal organism of typhoid is the bacterium *Salmonella enterica serovar Typhi* (commonly referred to as *Salmonella Typhi*).

Symptoms of Typhoid:
- High fever (often spiking at 39–40°C)
- Weakness and fatigue
- Abdominal pain and discomfort
- Diarrhea or constipation
- Loss of appetite
- Rash (rose spots) on the abdomen
- Headaches and muscle aches
- In severe cases, it can lead to complications like intestinal perforation.

2. Pneumonia:
Pneumonia can be caused by various pathogens, including bacteria, viruses, and fungi. The most common bacterial causes are *Streptococcus pneumoniae* and *Haemophilus influenzae*.

Symptoms of Pneumonia:
- High fever and chills
- Cough with greenish or yellow sputum (sometimes blood-streaked)
- Shortness of breath and difficulty breathing
- Chest pain, often worsening with coughing or deep breathing
- Fatigue and general weakness
- Nausea, vomiting, or diarrhea in some cases.

> Quick Tip
>
> For typhoid, look out for fever and abdominal discomfort, and for pneumonia, focus on respiratory symptoms like cough, fever, and difficulty breathing.

7. Write an essay on Biotechnology.

Solution:

Essay on Biotechnology:

Biotechnology is a branch of science that involves the use of living organisms, cells, or biological systems to develop products and technologies aimed at improving human life and the environment. It is an interdisciplinary field that combines principles of biology, chemistry, physics, and engineering to solve problems related to health, agriculture, and industry. The major subfields of biotechnology include medical biotechnology, agricultural biotechnology, industrial biotechnology, and environmental biotechnology.

1. Medical Biotechnology:

Medical biotechnology focuses on the development of drugs, vaccines, diagnostic tools, and gene therapy. One of the significant achievements in this field is the development of recombinant DNA technology, which allows the creation of genetically engineered organisms to produce pharmaceuticals. For example, the production of insulin using genetically modified bacteria has revolutionized the treatment of diabetes. Moreover, biotechnology is used in personalized medicine, where therapies are tailored to the genetic makeup of individual patients.

2. Agricultural Biotechnology:

Agricultural biotechnology involves the use of genetic engineering, plant breeding, and molecular biology techniques to enhance crop yields, resistance to pests, and tolerance to environmental stresses. Genetically modified (GM) crops, such as Bt cotton and Roundup Ready soybeans, have been developed to provide higher productivity and reduce dependency on chemical pesticides. Additionally, biotechnology plays a crucial role in improving food security by developing crops with enhanced nutritional content, like Golden Rice, which is fortified with vitamin A.

3. Industrial Biotechnology:

Industrial biotechnology refers to the use of microorganisms, enzymes, and other biological agents in manufacturing processes. This includes the production of biofuels, biodegradable plastics, and enzymes used in various industries such as textiles, paper, and food processing. Biotechnology has led to the development of more sustainable manufacturing processes that are less harmful to the environment.

4. Environmental Biotechnology:

Environmental biotechnology focuses on the use of biological systems to reduce pollution and manage waste. Bioremediation, which uses microorganisms to clean up pollutants like oil spills or heavy metals, is a key application. Additionally, biotechnology is used in wastewater treatment, where bacteria break down organic waste, and in the development of green technologies such as bioenergy and waste-to-energy solutions.

Conclusion:

Biotechnology has made significant advancements in various fields, improving human health, increasing agricultural productivity, and promoting environmental sustainability. However, it also raises ethical and environmental concerns, especially regarding the use of genetically modified

organisms (GMOs) and potential ecological impacts. Despite these challenges, biotechnology continues to be a rapidly evolving field that holds great promise for addressing global challenges such as disease, food security, and climate change.

Quick Tip

Biotechnology combines biology, chemistry, and engineering to solve problems in medicine, agriculture, industry, and the environment.

OR

(b) Describe the process of microsporogenesis in Angiosperms with diagrams.

Solution:

Microsporogenesis is the process by which microspores are formed from the microspore mother cells (or pollen mother cells) in the anthers of angiosperms (flowering plants). It is the first step in the formation of male gametes (pollen grains) that participate in fertilization. The process of microsporogenesis occurs in the following stages:

1. Formation of Microspore Mother Cells (PMC):

The anthers of a flower consist of four microsporangia. Each microsporangium contains diploid microspore mother cells (PMC). These cells are derived from the sporogenous tissue, which is composed of diploid cells.

2. Meiosis (Reduction Division):

Each microspore mother cell undergoes meiosis, a type of cell division that reduces the chromosome number by half. Meiosis consists of two divisions: - **Meiosis I**: The diploid PMC undergoes the first meiotic division, resulting in two haploid cells. - **Meiosis II**: The two haploid cells divide again, producing four haploid microspores. These microspores are the result of reduction division, and each contains a single set of chromosomes (n).

3. Formation of Microspores:

After meiosis, the four microspores are formed, each surrounded by a thin cell wall. These microspores are haploid, meaning they carry half the genetic material of the parent cell.

4. Development of Pollen Grain (Microgametogenesis):

The microspores then undergo mitotic divisions to form the pollen grains. Each microspore develops into a pollen grain, which consists of two cells: a large vegetative cell and a smaller generative cell. The vegetative cell forms the tube cell, which eventually forms the pollen tube during fertilization, and the generative cell divides further to form two male gametes (sperm cells).

Quick Tip

Microsporogenesis produces four haploid microspores from one diploid microspore mother cell through meiosis, and each microspore forms a pollen grain that contains two cells: the vegetative and generative cells.

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8. What do you mean by Biodiversity? Describe its different methods of conservation in detail.

Solution:
Biodiversity refers to the variety and variability of life on Earth. It includes the diversity of species, ecosystems, and genetic variations within species. Biodiversity is essential for ecosystem stability, resilience, and providing ecosystem services such as clean air, water, food, and climate regulation.

Methods of Conservation of Biodiversity:
There are two main approaches to biodiversity conservation: **in-situ conservation** and **ex-situ conservation**. Both methods aim to protect the natural habitat and the species within it.

1. In-Situ Conservation:
In-situ conservation refers to the conservation of species in their natural habitats. It is the most effective way of protecting biodiversity. The following are key methods of in-situ conservation:

- **Protected Areas**: National parks, wildlife sanctuaries, and biosphere reserves are designated areas where human activities are restricted to protect wildlife and plant species.
- **Wildlife Corridors**: These are pathways that connect isolated protected areas, allowing species to migrate, interbreed, and access new habitats.
- **Sacred Groves**: Areas protected and preserved by local communities due to cultural or religious beliefs.
- **Restoration of Habitats**: Active restoration of damaged ecosystems, such as reforestation and wetland restoration, to revive biodiversity.

2. Ex-Situ Conservation:
Ex-situ conservation involves the preservation of species outside their natural habitats. This method is used for species that are at risk of extinction or whose habitats are destroyed. Some common ex-situ conservation methods are:

- **Botanical Gardens**: Institutions that conserve plant species, especially rare and endangered ones, outside their natural environment.
- **Zoological Parks**: Zoos that house animals in artificial environments, where they can be bred, studied, and sometimes reintroduced into the wild.
- **Seed Banks**: Facilities where seeds of various plant species are stored for future use, such as in case of habitat destruction.
- **Cryopreservation**: The preservation of genetic material, such as sperm, eggs, and seeds, at

extremely low temperatures for future research and breeding.

> Quick Tip
>
> In-situ conservation preserves biodiversity within natural habitats, while ex-situ conservation preserves species outside their natural environment.

OR
(b) Describe the structure of human sperm and ovum with diagrams.

Solution:
The structure of human sperm and ovum are essential for reproduction, and they are specifically adapted to perform their respective roles in fertilization.

1. Structure of Human Sperm:
Human sperm consists of three main parts: the head, the midpiece, and the tail (flagellum). Each part plays a crucial role in the sperm's function.

- **Head**: The head contains the nucleus, which holds the sperm's genetic material (23 chromosomes). The tip of the head, called the acrosome, contains enzymes that help the sperm penetrate the ovum during fertilization.
- **Midpiece**: The midpiece is rich in mitochondria, which provide energy for the sperm's movement. It connects the head to the tail.
- **Tail (Flagellum)**: The tail is responsible for the sperm's motility, allowing it to swim toward the egg. It moves in a whip-like fashion to propel the sperm.

2. Structure of Human Ovum (Egg Cell):
The human ovum is a large, spherical cell and contains the female's genetic material (23 chromosomes). It consists of the following parts:

- **Nucleus**: The nucleus contains the egg's genetic material, which combines with the sperm's genetic material during fertilization to form a zygote.
- **Cytoplasm**: The cytoplasm surrounds the nucleus and contains nutrients and organelles needed for the early stages of embryo development.
- **Zona Pellucida**: A glycoprotein layer surrounding the egg that plays a role in fertilization by allowing only one sperm to enter the egg.
- **Corona Radiata**: A layer of cells that surrounds the zona pellucida and provides the egg with protection and nutrients.

Quick Tip

The sperm is designed for mobility and genetic delivery, while the ovum is designed for nutrient storage and genetic contribution to fertilization.

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9. Describe the dihybrid cross with the help of checker board.

Solution:
A dihybrid cross is a genetic cross between two individuals that involves the inheritance of two traits. The two traits are controlled by two different genes, each with two alleles (dominant and recessive). This cross follows the principles of Mendelian inheritance. In order to predict the genetic outcome of a dihybrid cross, we use a Punnett square or checkerboard. Here's a step-by-step explanation:

Step 1: Assigning Alleles and Parental Genotypes
Consider two traits:
- Trait 1: Seed color (Yellow - Y, Green - y)
- Trait 2: Seed shape (Round - R, Wrinkled - r)

Assume both parents are heterozygous for both traits (YyRr).

Step 2: Gamete Formation
Each parent can produce four different types of gametes, formed by the independent assortment of alleles:
- Parent 1 (YyRr) can produce the following gametes: YR, Yr, yR, yr
- Parent 2 (YyRr) can produce the same gametes: YR, Yr, yR, yr

Step 3: Setting up the Punnett Square (Checkerboard)
Now, set up a 4x4 Punnett square, where each gamete from one parent is placed along the top and the other parent's gametes along the side. The results are as follows:

	YR	Yr	yR	yr
YR	$YYRR$	$YYRr$	$YyRR$	$YyRr$
Yr	$YYRr$	$YYrr$	$YyRr$	$Yyrr$
yR	$YyRR$	$YyRr$	$yyRR$	$yyRr$
yr	$YyRr$	$Yyrr$	$yyRr$	$yyrr$

Step 4: Genotypic and Phenotypic Ratios
- Genotypic ratio: 1 YYRR : 2 YyRR : 2 YYRr : 4 YyRr : 1 YYrr : 2 Yyrr : 1 yyRR : 2 yyRr : 1 yyrr.
- Phenotypic ratio: 9 Yellow and Round : 3 Yellow and Wrinkled : 3 Green and Round : 1 Green and Wrinkled.

Quick Tip

In a dihybrid cross, the phenotypic ratio for two heterozygous parents is 9:3:3:1.

OR

(b) What do you mean by adaptations? Describe various adaptations in living organisms with suitable examples.

Solution:

Adaptations are specialized characteristics or behaviors of an organism that increase its chances of survival and reproduction in its environment. Adaptations can be structural, physiological, or behavioral, and they are shaped by natural selection. The following are different types of adaptations:

1. Structural Adaptations:

These adaptations involve physical features of an organism that help it survive in its environment. Examples include:

- **Camouflage**: Animals like chameleons and peppered moths have developed the ability to blend in with their surroundings to avoid predators.
- **Mimicry**: Some harmless organisms, like the viceroy butterfly, mimic the appearance of harmful species (such as the monarch butterfly) to avoid being eaten.
- **Body Shape and Size**: The long neck of the giraffe allows it to feed on tree leaves high above the ground, and the streamlined body of a fish helps it swim efficiently.

2. Physiological Adaptations:

These adaptations involve changes in the internal processes of an organism. Examples include:

- **Thermoregulation**: Animals like polar bears have thick fur and a layer of fat to insulate them from the cold, while desert animals like camels can tolerate extreme heat and conserve water.
- **Photosynthesis**: Plants have adapted to different environments by modifying their photosynthesis process. For example, cactus plants in deserts use CAM (Crassulacean Acid Metabolism) to minimize water loss.

3. Behavioral Adaptations:

These adaptations involve the actions or behaviors of organisms to survive and reproduce. Examples include:

- **Migration**: Birds migrate to warmer climates during winter to find food and suitable breeding grounds.
- **Hibernation**: Bears and some other animals hibernate during the winter to conserve energy when food is scarce.
- **Nocturnal Behavior**: Animals like owls and bats are active at night to avoid predators and

take advantage of cooler temperatures in hot environments.

Quick Tip

Adaptations are crucial for survival and reproduction, and they vary across species depending on their environment.
